6. Basic Economic Analysis

Introduction

From the discussion presented in the previous chapter, it is clear that it is necessary to determine a method for comparing alternatives in a standard manner. The alternatives being compared may have different costs of implementation. Given the same result in meeting the objective stated for control, it is desired that the best alternative be chosen. In the simplest case, this is done by selecting the lowest cost alternative.

However, the lowest cost alternative is not always easy to determine at first glance. The initial investment may differ between alternatives, and the operational costs may vary. The life of the equipment between two alternatives being considered or compared may also vary. Labor costs to operate and maintain the equipment as well as space requirements for each of the alternatives may be different. A method is needed to put these costs on an equal basis for comparison.

The first step in comparing alternatives is to determine those alternatives that meet the objective. If two alternatives meet a given objective, then these alternatives are comparable. If, on the other hand, one of the alternatives does not meet the objective, then it is not comparable to another alternative that does meet the objective. This fact emphasizes the need to state clearly the objectives of control. The objectives of control should be stated so that:

a. the desired result is specified,
b. the desired result can be measured,
c. the required conditions of performance are specified,
d. the method for determining whether the desired result has been reached is specified, and
e. the acceptable tolerances are identified.

Once the objective has been stated and clearly defined, the various alternatives can be compared to this objective to determine if the objective will be attained. If an alternative results in meeting the objective, it can be considered further. If the objective is not reached by implementation of an alternative, then this alternative can be eliminated from any further comparison.

Finally, after eliminating those alternatives that do not meet the objective, the remaining alternatives must be compared in some manner. An

economic analysis provides the best method for making such a comparison. This economic analysis considers the initial investment cost, the yearly operating cost, and the savings generated over existing methods. That alternative which results in the best economic advantage to the organization is the alternative that should be implemented.

In this chapter, two methods of economic analysis will be discussed; the N-year payback method and the present-worth approach. Variations of these methods, as well as other methods, have been developed and will provide equally useful analysis techniques. However, it is felt that, for the most part, these two methods are sufficient for conducting an economic analysis of capital investments.

The N-Year Payback

The N-Year payback approach for comparing capital investments is perhaps the simplest technique to use. This approach is particularly useful when it is necessary to compare a new method of control to an existing method of control. Using the N-Year payback, the costs that are incurred for the present method of control are compared to the projected costs for the new method being considered. The difference in cost between the methods, i.e., the savings, is then compared to the required investment to implement the new method of control. This comparison results in a payback for the new method being considered. When using this approach, the assumption is made that the savings are equal for each year of life of the new investment. The payback formula can be stated as

$$(8.6.1) \quad \text{Payback } (P) = \frac{I}{S}$$

where

I = the capital investment cost to obtain the savings
S = the savings generated by the new method as compared to the existing method
P = the number of years required to pay back the investment

In using the N-year payback method, it is necessary to choose a cutoff payback period. This varies to some degree upon the type of investment. Generally, three to five years is used for equipment; whereas, plant investments may use a ten-year or greater payback period. Most organizations have specific desired payback periods for different types of investments. To illustrate the N-year payback, the following example is presented.

Example Problem

A plant operates with an intermittently cleaned bag house to remove particulates from the exhausted air. The bag house requires cleaning daily, involving production shutdown for one hour and an operator at $12 per hour including fringe benefits. The bag house was purchased one (1) year ago at an installed cost of $50,000. A newly-designed bag house can

be obtained for $160,000 installed. The new design will not require shutdown for cleaning since it is cleaned continuously. The new design will operate at the same efficiency in removing particulates but will cost $50 per day more to operate. One hour's production has an estimated value of $300 profit to the firm. Assuming a three (3) year payback is required for investments of this type, answer the following questions.

a. If the plant is now operating to capacity (3 shifts), 365 days per year, and if additional capacity can be sold, should the investment be made?
b. If the plant is operating two shifts per day on a 365 day basis, should the investment be made?
c. Is any other alternative available, and if so, is this alternative a better investment than the new equipment?

Using the N-year payback method, the first step is to determine the investment involved. In this case, the investment is $160,000. The reader should be cautioned not to consider the $50,000 spent on the existing equipment since this represents a "sunk" cost. The concept of "sunk" costs involves costs or investments that have been made in the past. These costs do not in any way effect the determination as to whether to implement a given new alternative.

The savings to be generated in the first case where the plant is operating at full capacity and where additional capacity provided by removing the need for production downtime can be sold is as follows:

Savings

a. Labor Cost = $12 per hour x 365 days = $4,380
Production Revenue = $300 per hour x 365 days = $109,500
The costs of operating the new equipment must be determined since these costs will result in a lower net savings to the firm. These costs are:

Operating Cost = $50 per day x 365 days = $18,250

The savings are then:

Net savings = savings - cost
Net savings = ($4,380 + 109,500) - $18,250
Net savings = $95,630

The payback can be calculated by the formula:

Payback = I/S
Payback = $160,000/95,630
Payback = 1.67 years

b. In the second case, the additional capacity is not needed. The investment remains the same at $160,000. However, the savings now involve only the labor cost since no production revenue will be obtained by eliminating the need for downtime. The cost of operating the new

equipment is still $18,250. In this case, the net savings is--$13,870. Obviously, the investment under this situation is a losing proposition.

c. One alternative that was not specifically stated in the problem but which can be considered is to install a second intermittent-type bag house in parallel with the first. The second bag house can be operated to clean the air while the first bag house is shut down for cleaning. This will not require a shutdown of the production facility.

To compare this alternative with the existing situation and assuming that the installed cost has not changed over the last year, the following is developed:

Investment = $50,000

Savings:

Labor Cost = Same as present method (alternate day cleaning)
Production Revenue = $300 per hour x 365 days
Production Revenue = $109,500

$$\text{Payback} = \frac{\$50,000}{\$109,500}$$

Payback = 0.46 years

This is clearly the best economic alternative. This situation points out the necessity to consider other alternatives to make sure that the best alternative will be chosen. If only the obvious alternative had been considered and chosen, then a significant opportunity cost would have been incurred by the company.

Economic Analysis Using the Present Worth Approach

The simple method of the N-year payback does not always provide the best approach for conducting an economic analysis. The cost and savings flow may vary from year to year between the alternatives being compared. In addition, unequal life cycles may be present. In order to consider these factors, it is necessary to look at the investment in another way.

One method is to consider the average savings that are generated. However, using this approach does not consider the fact that a dollar earned today is worth more than a dollar earned tomorrow. Let's look at this concept a little further.

If a firm earns a dollar today, that dollar can be invested in such a manner that additional money can be made. For example, consider the simple case of putting a dollar in a savings bank. If the rate of interest earned in a savings bank is 6%, then the dollar will be worth $1.06 at the end of the year. A dollar earned next year cannot be invested until it has been earned and, therefore, is still only worth a dollar. Thus, the dollar earned this year is worth six cents more than the dollar earned next year. It is on this concept that the present worth approach is based.

The objective of the present worth approach is to put all costs and savings in terms of current dollars. This is done by using a compound interest rate to discount the money back to its equivalent worth in dollars today. Inflation is not considered since it is assumed that it will equally affect costs and profit. The rate that is used to discount the money to its present worth is determined by either looking at the cost to borrow capital or determining an acceptable rate of return on investment from the profit return rate the organization earns.

Let's look at a simple example to see how the present worth approach works. Consider $10 invested today. How much more is this $10 worth than $10 earned in one year or $10 earned in two years, if it is assumed that a return of 5% can be obtained on any investments made?

Ten dollars will yield $10.50 in one year at 5% interest. This is shown by the following:

$$\text{Value} = \$10 + 10(.05) = \$10.50$$

This can be generalized for any interest rate and investment by:

$$(8.6.2)\quad \text{Value} = (1 + i)1$$

Ten Dollars in two years can earn:

$$\text{Value} = 10(1 + .05) + 10(1 + .05)(.05)$$

Generalizing from this, we obtain the following:

$$\begin{aligned}\text{Value} &= 1(1 + i) + 1(1 + i)(i)\\ \text{Value} &= 1[1 + i + i + i^2]\\ &= 1[1 + 2i + i^2]\\ &= 1(1 + i)^2\end{aligned}$$

This can be generalized for N years in the following way:

$$(8.6.3)\quad \text{Value} = 1(1 + i)^n$$

Now, the question must be asked, "How much money would have to be earned in the future to be worth the same as money earned today?" This can be stated as:

$$(8.6.4)\quad \text{Future Earnings} = 1(1 + i)^n$$

Future earnings are then worth what value today? Transposing the formula above, the following is obtained:

$$(8.6.5)\quad 1 = \frac{\text{Future Earnings}}{(1 + i)^n}$$

This is the basis for the present-worth approach. The formula for determining the present worth of a dollar earned or saved in the future is as follows:

$$(8.6.6)\quad PW_i = S \times \frac{1}{(1 + i)^n}$$

where

PW_i = the present worth of a future savings at a rate of interest
S = the savings earned at some future point in time
i = interest rate
n = years until savings are earned

Consider the following example which illustrates the use of the present worth approach. A series of savings are earned at a rate of return of 8 percent.

At the End of Year	Savings($)
1	400
2	200
3	500
4	300

Using the present worth approach, the following calculation can be made:

Year	Savings($)	x	Factor	=	Present Worth($)
1	400		$1/(1 + 0.08)$		370.37
2	200		$1/(1 + 0.08)^2$		171.47
3	500		$1/(1 + 0.08)^3$		396.42
4	300		$1/(1 + 0.08)^4$		220.51
					$1,159.27

To determine the rate of return on an investment, the present worth of the investment at a given rate of return minus the present worth of the savings earned at this same return must be equal to 0. In the above case, to earn a rate of return of 8 percent, the investment today must be equal to $1,159.27 for the investment and present worth of savings to be equal. Perhaps this method can be better shown by considering another problem involving present worth.

Example

A manufacturer has been using a dilution ventilation system for control of nuisance dusts in a work area. However, with the cost of fuel rising, a local exhaust system is being considered as a replacemen for the existing

system. It is estimated that the cost of tempering makeup air will be reduced to one-fourth of the present cost of $50,000 annually using the local exhaust system. Also it is expected that the present dilution system will require a major overhaul at the end of 5 years at a cost of $50,000. The investment cost for the new system is $300,000. If the desired rate of return for the firm is 6%, should the investment be recommended? Assume a 10-year life for the project.

Solution

In each year, the savings obtained by making the investment will be as follows :

Heating + power = Total savings
$50,000 x 3/4 + $8,000 = $45,500

In year 5, the additional overhaul costs of $50,000 will also be saved. The resulting savings and present worth at 6% are shown in the following table.

Year	Savings($)	Present Worth Factor $[1/(1 + i)^n]$	Present Worth($) (Savings)
1	45,500	.94339	42,925
2	45,500	.88999	40,495
3	45,500	.83961	38,203
4	45,500	.79209	36,040
5	45,500	.74725	34,000
6	95,500	.70496	67,324
7	45,500	.66505	30,260
8	45,500	.62741	28,547
9	45,500	.59189	26,931
10	45,500	.55839	25,407
			$370,132

Since the following is the case:

$PW_{.06}$ (Savings) > $PW_{.06}$ (Investment)

That is, $370,132 > $300,000, then the rate of return criterion is exceeded and the investment should be recommended.

If it were desired to determine the actual rate of return that is obtained from this investment, it would be necessary to make a trial and error calculation using different rates of return until the present worth of the savings is equal to the present worth of the investment. The rate at which the two values are equal is the rate of return for the investment.

Effects of Taxes on Economic Analysis

Taxes play an important part in the economic decisions that are made in industry. Since taxes affect profits and are paid after expenses and depreciation are deducted, they can have an appreciable effect on an economic analysis, given that certain conditions exist involving the investments. The major factor that can affect the tax structure is the use of depreciation. Depreciation is the method for charging the cost of equipment purchased over the useful life of the equipment. Expenses are charged as they are incurred since they have no life beyond the present year.

The amount which can be depreciated for a capital investment is calculated as the difference between the capital investment minus any salvage value that the equipment has at the end of its useful life. The depreciation can then be obtained in a number of ways. Two methods that are commonly used are the straight line depreciation method and the declining balance method. In the straight line depreciation method, the difference between the capital investment and the salvage value is distributed equally over each year in the life of the equipment. Thus, if an investment has a capital cost of $10,000 with a $2,000 salvage value and an expected life of 10 years, the depreciation chargeable each year would be $800.

In the declining balance method, an attempt is made to depreciate the larger amount of money during the early life of the equipment. This is done based upon the principle that a dollar today is worth more than a dollar tomorrow and any charges made will reduce the tax liability of the corporation, thus, in effect earning dollars for the corporation. The formula for calculating depreciation based upon the declining balance method is

$$(8.6.7) \quad D = \frac{2}{n} + (1-2/n)^{j-1} \quad \text{for } j = 1, \ldots, n$$

where

D = the depreciation for year j

$\frac{2}{n}$ = the maximum allowable percentage depreciation, n = number of years life of the investment

I = investment

j = year depreciation

This formula includes the salvage value as a part of the original investment since this money will not be returned until year n. That is, the salvage value is $I(1 - s/n)^n$ in terms of present dollars.

Using this depreciation method for a $5,000 investment with an expected life of 5 years and a salvage value of $500, the depreciation and tax write-off at a 50% tax are as follows:

Year	Depreciation($)	Tax Write-Off($)
1	2,000	1,000
2	1,200	600
3	720	360
4	432	216
5	148*	74

*Since you cannot depreciate more than (I - SV), you must use straight line depreciation over the last two years.

Other depreciation methods have been approved for use by the Internal Revenue Service. One of the major of these is the sum-of-the-years-digit-method. However for the purposes of this discussion, other methods will not be discussed. The reader may refer to a basic accounting book for a further explanation of depreciation methods.

In order to stimulate the economy, over the past few years the Federal government laws have given an investment credit for capital expenditures. This investment credit can be charged against the current year's taxes and is an incentive for purchasing equipment.

Taxes can affect the decision to make an investment. The effect of taxes depends upon the flow of capital versus expense dollars. The investment credit can have an effect on the decision. The tax laws are constantly changing; and, as a result, it is advisable for the industrial hygiene engineer to consult an expert in the area of tax laws to interpret the current law and its effects on an investment.

Problem

What will be the effect of taxes on the problem presented in the previous example if the desired after-tax rate of return is 6% and if the $50,000 overhaul cannot be expensed? Use a straight line depreciation for the 10-year life of the local exhaust system, a 7.5% investment credit, and a tax rate of 50%. Assume no salvage value for the local exhaust system after its life and 5-year life on the $50,000 overhaul.

Year	After Tax Savings Depreciation ($)	x	Present Worth$_{.06}$	= Present Worth$_{.06}$ (S)($)
1	(22,750 + 15,000)		.94339	35,613
2	(22,750 + 15,000)		.88999	33,597
3	(22,750 + 15,000)		.83961	31,695
4	(22,750 + 15,000)		.79209	29,901
5	(22,750 + 15,000)		.74725	28,209
*6	(22,750 + 6,250)		.70496	20,444
7	(22,750 + 10,000)		.66505	21,780
8	(22,750 + 10,000)		.62741	20,548
9	(22,750 + 10,000)		.59189	19,384
10	(22,750 + 10,000)		.55839	18,287
				$259,458

*Includes 7.5% investment credit on overhaul in year 6 and the difference in depreciation between the overhaul and the new equipment for the remaining years.

Since

$PW_{.06}$ (Savings) < $PW_{.06}$ (Investment - .075 Investment), the investment is not recommended.

In the above example, it is obvious that a different present worth would be obtained if a declining balance depreciation were used since this would load the savings in the early life of the equipment. The student is encouraged to work out the problem in this manner to observe the effect.

Summary

In this chapter, the need for economic analysis of various alternative investments has been presented. Two methods for analyzing investments have been discussed: the N-year payback method and the present worth approach. The N-year payback method is applicable where the savings are equal over the life of the equipment, while the present worth approach takes into consideration varying rates of savings and expense. In addition, the effect of taxes on an investment may be of concern. Of particular importance is the distribution of costs in terms of capital investment which must be depreciated and expenses which can be charged in the year when they are incurred. Depending upon the flow of investments versus expenses, taxes can have a major effect on an investment decision.